

Descriptive Statistics

Q1: Understanding Central Tendency (Easy) A bakery tracks the daily sales of muffins (in dozens) over a week: [10, 12, 11, 15, 14, 13, 12]. What is the most representative value of their weekly sales, and why?

Sales(in dozens) = [10, 11, 12, 12, 13, 14, 15]

Mean = $(10+12+11+15+14+13+12) / 7 = 12.42$

Median = 12

Mode = 12

The most representative weekly sales value is **12 dozen** muffins per day, since it is both the median and mode and is very close to the mean, making it a good summary of the bakery's typical daily sales.

Q2: Mean in Real Life (Easy) A teacher records the marks of her students in a short quiz: [12, 15, 14, 16, 18, 20, 19]. What is the mean score, and what does it tell us about the class's performance?

Mean score = $(12+15+14+16+18+20+19) / 7 = 16.28$

The mean score of **16.28 marks** indicates that the class performed, on average, around **16 marks** on the quiz. It provides a single value that summarizes the overall performance of the class.

Q3: Mode in Real Life (Easy) A store records the shoe sizes sold in one day: [7, 8, 9, 8, 8, 10, 7, 9]. What is the mode, and why is this information useful for the store manager?

Shoe sizes = 7,7,8,8,8,9,9,10

Mode = 8

Size **8** is the most frequently sold shoe size, so the store manager can use this information to ensure sufficient stock and improve inventory management.

Q4: Median in Real Life (Medium) A car dealer notes the prices of used cars: [\$8,000, \$9,500, \$10,200, \$11,000, \$50,000]. Why is the median a better measure than the mean in this case? Calculate the median.

Prices of used cars = [\$8,000, \$9,500, \$10,200, \$11,000, \$50,000]

Median = \$10,200

Mean = $(8000 + 9500 + 10200 + 11000 + 50000) / 5 = \17740

The mean is **\$17,740**, which is much higher than most of the car prices. This happens because the **\$50,000 car is an outlier** (an unusually expensive car) that pulls the mean upward.

The **median is not affected by extreme values**, so it better represents the typical price of the used cars being sold.

Q5: Dispersion Introduction (Medium) A student times how long it takes to finish a puzzle each day: [25, 30, 27, 35, 40]. What does the range tell us about the variation in the student's puzzle-solving time?

$$\begin{aligned}\text{Range} &= \text{max_value} - \text{min_value} \\ &= 40 - 25 = 15\end{aligned}$$

The range here shows that 15 min of time difference from the student who finished 1st and the student who finished last.

A larger range would indicate greater variation, while a smaller range would indicate more consistent puzzle-solving times.

Q6: Range in Action (Medium) A farmer records the weekly weight of harvested apples (kg): [100, 105, 98, 110, 120]. Find the range. How can this help the farmer in planning his packaging?

$$\text{Range} = 120 - 98 = 22\text{kg}$$

The range shows that the harvest can vary by **22 kg** from week to week. This helps the farmer prepare appropriate packaging, storage, and transportation arrangements to handle fluctuations in production.

Q7: Variance for Decision-Making (Medium) Two delivery companies track delivery delays (in minutes). Company A: variance = 6 Company B: variance = 15 Which company is more consistent, and why?

Since , $6 < 15$

Company A has lower variance so Company A is more consistent. Because its delivery delays vary less from the average delay. Customers can expect delivery times to be more predictable with Company A, whereas Company B's delays fluctuate more from one delivery to another.

Q8: Standard Deviation in Context (Hard) A finance student compares the daily price fluctuations of two cryptocurrencies. Coin A: standard deviation = \$30 Coin B: standard deviation = \$120 Which coin is riskier to invest in, and why?

Coin A : Standard Deviation = \$30

Coin B : Standard Deviation = \$120

Coin B is riskier because it has a higher standard deviation (\$120 vs. \$30). A higher standard deviation indicates greater price volatility and uncertainty, making the investment more risky.

Q9: Combining Measures (Hard) A family records their monthly electricity usage (in kWh): [400, 420, 390, 450, 410]. Find the mean and standard deviation. What do these values together tell you about the family's energy use pattern?

Monthly Electricity usage(kWh) : [400, 420, 390, 450, 410]

Mean = (400 + 420 + 390 + 450 + 410) / 5 = 414 kWh

Standard Deviation

Using the population standard deviation formula:

$$\sigma^2 = \sum (x - \mu)^2 / N$$

Value (x)	$x - 414$	$(x - 414)^2$
400	-14	196
420	6	36
390	-24	576
450	36	1296
410	-4	16
$\sum (x - \mu)^2 = 196 + 36 + 576 + 1296 + 16 = 2120$		

Variance: $\sigma^2 = 2120 / 5 = 424$

Standard Deviation: $\sigma = 20.6$ kWh

Mean electricity usage = 414 kWh

Standard deviation ≈ 20.6 kWh

The family has a fairly consistent pattern of electricity consumption, with only moderate month-to-month variation around their average usage of 414 kWh.

Q10: Practical Application (Hard) A basketball player's points in 8 games are recorded: [15, 18, 20, 22, 25, 17, 19, 21]. Find the mean, median, mode, range, and standard deviation. What insights can these measures provide about the player's scoring performance?

[15, 17, 18, 19, 20, 21, 22, 25]

$$\text{Mean} = (15+17+18+19+20+21+22+25) / 8 = 19.625$$

$$\text{Median} = 19 + 20 / 2 = 19.5$$

Mode = none

$$\text{Range} = 25 - 15 = 10$$

Std dev :

Using the population standard deviation:

$$\text{Mean}(\bar{x}) = 19.625$$

Score	$X - \bar{x}$	$(x - \bar{x})^2$
15	-4.625	21.3906
18	-1.625	2.6406
20	0.375	0.1406
22	2.375	5.6406
25	5.375	28.8906
17	-2.625	6.8906
19	-0.625	0.3906
21	1.375	1.8906

$$\sum (x - \bar{x})^2 = 67.875$$

$$\text{Variance: } \sigma^2 = 67.875 / 8 = 8.484$$

$$\text{Standard deviation: } \sigma = \text{SQRT}(8.484375) \approx 2.91$$

The player is a consistent scorer, typically scoring around **20 points** per game. The low standard deviation and the closeness of the mean and median indicate stable and reliable performance with no extreme highs or lows.